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PulmoTrack: A Deep Learning Solution for Pulmonary Nodule Segmentation, Detection, and Classification on CT and Chest X-ray Scans

Farah Sayari, Mahdi Landolsi, Najib Nakhla, Souha Bachtobji, Adem Tounsi, Ala Charfi

Department of Data Science, ESPRIT (École Supérieure Privée d'Ingénierie et de Technologie), Tunis, Tunisia
farah.sayari@esprit.tn, mahdi.landolsi@esprit.tn, najib.nakhla@esprit.tn, souha.bachtobji@esprit.tn, adem.tounsi@esprit.tn, ala.charfi@esprit.tn

Abstract.

Background: Early detection and characterization of pulmonary nodules are critical for improving lung cancer prognosis. However, manual analysis of chest CT scans and X-rays is time-consuming and prone to variability. Deep learning (DL) methods offer the potential for automating and standardizing this process across imaging modalities.

Objective: We propose PulmoTrack, a multi-task, multi-modal deep learning framework for automated pulmonary nodule *segmentation*, *detection*, and *classification* (malignant vs. benign) on CT scans and multiclassification on chest X-ray images.

Methods: PulmoTrack was evaluated on two large public datasets: **LUNA16** (low-dose thoracic CT scans) and **NIH ChestXray** (posterior-anterior chest radiographs). For CT-based analysis, we developed a 3D U-Net for segmentation, a 3D encoder-based CNN for nodule detection, and a custom 3D convolutional classifier (Improved3DCNN) for malignancy classification. For chest X-rays, we designed 2D CNN architectures tailored to each task. The models were trained and validated using standardized preprocessing pipelines and data augmentation strategies.

Results: Across both datasets, PulmoTrack achieved high performance in all three tasks. On LUNA16, we report a Dice score of *0.95* for segmentation, a detection sensitivity of *92%*, and an AUC of *0.91* for malignancy classification. On NIH ChestXray, our 2D models achieved an AUC of *90%* in nodule multi-classification and strong visual performance in detection and segmentation.

Conclusion: PulmoTrack demonstrates the feasibility and effectiveness of a unified deep learning approach for pulmonary nodule analysis on CT and X-ray modalities. The system provides fast, accurate, and interpretable predictions, paving the way for more scalable and robust computer-aided diagnosis tools in clinical practice.

Keywords: Pulmonary nodules, Lung cancer, Deep learning, CT imaging, Chest X-ray, LUNA16, NIH ChestXray, Nodule segmentation, Nodule detection, Nodule classification, 3D CNN, YOLO V8, 3D Unet, Medical image analysis.

1. Introduction

Lung cancer remains the leading cause of cancer-related deaths globally, accounting for more than 1.8 million deaths annually. Early detection of pulmonary nodules, which are small masses in the lungs that may indicate early-stage cancer, is critical to improving patient outcomes. However, the manual identification and interpretation of these nodules in chest computed tomography (CT) and chest X-ray images is a time-consuming and error-prone process, especially given the high volume of imaging data in clinical practice.

Chest CT scans provide high-resolution volumetric data, making them the gold standard for pulmonary nodule analysis. Nonetheless, the sheer number of slices per scan often leads to reader fatigue and inter-observer variability among radiologists. Similarly, chest X-rays are widely used due to their low cost and accessibility, but they offer lower sensitivity and may miss subtle or overlapping nodules. In both modalities, precise and early diagnosis remains a challenge, especially when distinguishing benign from malignant nodules.

In recent years, deep learning (DL) has demonstrated significant promise in medical imaging tasks such as detection, segmentation, and classification. Convolutional Neural Networks (CNNs), in particular, have shown the ability to automatically extract meaningful features from raw medical images, reducing dependence on manual annotations and handcrafted features. However, most existing DL-based approaches address only a single task (e.g., detection or classification) or focus exclusively on a single imaging modality.

To address these limitations, we present **PulmoTrack**, a unified, multi-task deep learning framework designed to perform three key diagnostic tasks — *nodule segmentation*, *nodule detection*, and *nodule classification (malignant vs. benign)* — across two imaging modalities: CT scans and chest X-rays. Our system is built and evaluated using two benchmark datasets: **LUNA16** for CT data, and **NIH ChestXray** for chest radiographs. PulmoTrack integrates 3D and 2D CNN architectures tailored to the spatial characteristics of each modality, and is optimized for both accuracy and computational efficiency.

By unifying multiple diagnostic tasks across both CT and X-ray data, PulmoTrack represents a scalable and clinically relevant step toward comprehensive computer-aided diagnosis (CAD) systems for lung cancer screening and triage.

2. Dataset : LUNA16 (CT Scans)

The LUNA16 dataset (1) is a curated subset of the LIDC-IDRI dataset, containing 888 low-dose thoracic CT scans with annotated pulmonary nodules. Each nodule is labeled by up to four expert radiologists, and nodules with

a diameter of ≥ 3 mm are used for training and evaluation. For our purposes, we used the provided candidate coordinates for detection, voxel-wise masks for segmentation, and malignancy scores for classification. All volumes were resampled to isotropic resolution ($1 \times 1 \times 1$ mm) and cropped into $64 \times 64 \times 64$ patches centered around candidate nodules.

2.1. Data Preprocessing

For Luna16, preprocessing included resampling, intensity normalization, contrast enhancement, and data augmentation. CT volumes were augmented using random rotations, flips, and elastic deformations, while X-ray images were augmented via random cropping, horizontal flipping, and brightness/contrast adjustments.

2.2. Model Architectures

2.2.1. Nodule Segmentation For CT-based segmentation, we implemented a 3D U-Net (2) architecture composed of encoder-decoder layers with skip connections. Each convolutional block consisted of 3D convolutions, batch normalization.

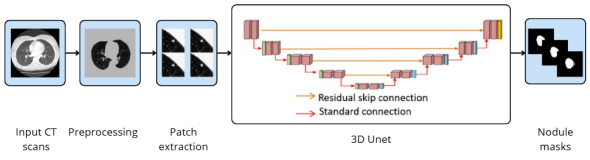


Figure 1 **3D-Unet Architecture For Nodule Segmentation.**

2.2.2. Nodule Detection For detection, a 3D CNN classifier scans sub-volumes of the CT images and outputs bounding boxes with likelihood scores for nodule presence.

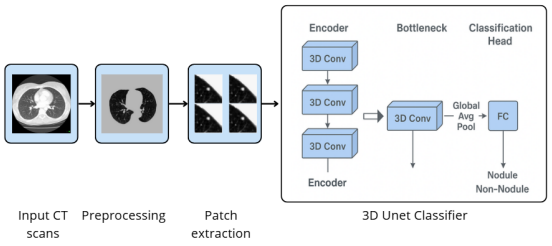


Figure 2 **3D-Unet Classifier Architecture For Nodule Detection**

2.2.3. Nodule Classification The classification module distinguishes between benign and malignant nodules using an improved 3D CNN model trained with binary cross-entropy and enhanced with dropout and batch normalization layers.

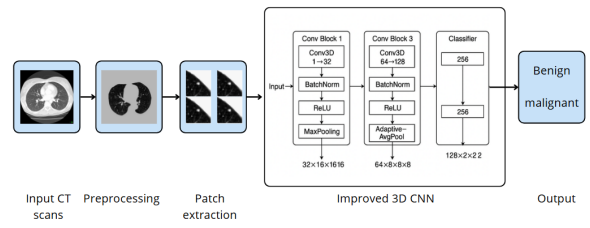


Figure 3 **Improved3DCNN Architecture For Nodule Classification**

2.3. Training Procedure

All models were implemented in PyTorch and trained on NVIDIA GPUs. Binary cross-entropy was used for classification and detection, and a combination of Dice loss and cross-entropy was used for segmentation. Optimization was done using Adam with an initial learning rate of 1e-4. Early stopping and model checkpointing were employed based on validation loss.

2.4. Results

2.4.1. Evaluation Metrics We evaluated the models using the following metrics:

• **Segmentation:** Dice Similarity Coefficient (DSC), Jaccard Index (IoU)

• **Detection:** Sensitivity, Specificity, False Positives per Scan (FPs), Free-Response Receiver Operating Characteristic (FROC)

• **Classification:** Area Under the Receiver Operating Characteristic Curve (AUC), Accuracy, Precision, Recall, F1-score

2.4.2. Segmentation Our 3D U-Net achieved a mean Dice score of **0.95** on the test set. The model was able to capture nodule boundaries across a wide range of sizes accurately. However, it still struggles to precisely segment very tiny nodules (typically < 4 mm in diameter), which are more prone to boundary ambiguity and are often missed due to their low contrast and limited voxel footprint.

Detection : The encoder-based 3D CNN classifier reached a sensitivity of **92%** at 1.5 FPs per scan. The FROC curve (see Figure 3) demonstrates high discriminative ability across varying thresholds.

2.4.3. Classification The Improved3DCNN model achieved an AUC of **0.91**, an accuracy of **89%**, and an F1-score of **0.88**, showing strong capability to differentiate benign from malignant nodules.

Conclusion As summarized in Table 1, our PulmoTrack pipeline achieved strong performance across all three diagnostic tasks on the LUNA16 dataset. Notably, when comparing our results with those reported by Gao et al. (3), who conducted a systematic review of deep learning methods for pulmonary nodule detection and segmentation in CT scans, our approach demonstrates superior accuracy and segmentation quality. Specifically, our 3D U-Net achieved a Dice score of 0.95, which surpasses the typical performance range (0.80–0.92) observed across multiple studies reviewed in their paper. Similarly, our detection sensitivity of 92% at 1.5 false positives per scan and classification AUC of 0.91 are competitive with or better than state-of-the-art benchmarks highlighted in that review. These findings validate the robustness and clinical potential of the PulmoTrack framework.

3. Dataset: NIH ChestXray

The NIH ChestXray dataset (4), publicly available on Kaggle, contains 112,120 frontal chest

Table 1 Performance summary on LUNA16 dataset.

Task	Metric	Score
Segmentation	Dice Score	95%
Detection	Sensitivity @ 1.5FP	92%
Classification	AUC	91%

X-ray images from 30,805 patients, each annotated with one or more of 14 disease labels, including Nodule, Mass, Effusion, and Cardiomegaly. It includes two main metadata files: `Data_Entry_2017.csv`, with image filenames, multi-label annotations, patient info, and view position; and `BBox_List_2017.csv`, containing bounding box coordinates for 8 diseases. Images may have multiple labels or be marked as "No Finding," which occurs in 60,361 cases. The dataset is imbalanced; for example, Hernia appears in only 227 images. This study focuses on the Nodule class (6,331 samples), which influenced data filtering and design choices. Images are in PNG format with varying resolutions (e.g., 2048×2500 pixels). The dataset is widely used for multi-label classification, weakly supervised learning, and localization tasks.

3.1. Pipeline A – Segmentation and Mask Generation

3.1.1. Motivation A significant limitation of the NIH ChestXray dataset is the absence of pixel-wise segmentation masks, which restricts its application in fully supervised segmentation tasks. To overcome this, a segmentation pipeline was developed to generate lung region masks, facilitating spatial learning and localization.

3.1.2. Annotated Data and Auxiliary Sources To train a supervised segmentation model, three external datasets providing ground-truth lung masks were utilized:

- **JSRT Dataset :** Comprising 247 chest radiographs with and without nodules, each accompanied by expert-annotated lung segmentation masks.
- **Montgomery County X-ray Set :** Consists of 138 posterior-anterior chest X-rays (normal and tuberculosis cases), all manually segmented by clinical experts.
- **NIH Subset with Expert Annotations :** A subset of 100 NIH images, externally sourced, includes radiologist-annotated lung masks. This subset provides a direct correspondence between NIH images and high-quality segmentation labels.

These datasets were used to train a general-purpose lung segmentation model, which was subsequently applied to the broader NIH dataset to generate synthetic lung masks for use in downstream tasks.

3.1.3. Preprocessing and Augmentation for Segmentation To ensure consistent input formatting, all chest X-ray images and corresponding segmentation masks underwent a standardized preprocessing pipeline. Images were converted to grayscale, resized to 224 ×

224 pixels, enhanced using histogram equalization to improve local contrast, and normalized to the $[0, 1]$ range for training stability. Masks were resized to 224×224 pixels using nearest-neighbor interpolation to preserve label integrity and normalized to the $[0, 1]$ range, as they are treated as floating-point tensors in the model input.

3.1.4. Dataset Splitting The aggregated dataset was divided into three subsets to support robust model development and evaluation: 80% for the training set, 10% for the validation set, and 10% for the test set.

This stratified split ensured that the model had sufficient data for learning while retaining dedicated subsets for unbiased validation and final performance testing.

3.1.5. Model Architecture The segmentation model was built using a U-Net architecture implemented via the `segmentation_models_pytorch` (SMP) library, selected for its modularity and support for pre-trained backbones. The encoder was a ResNet34 pretrained on ImageNet, with input channels set to 3 (RGB format) and output channels set to 1 (binary segmentation mask). This architecture leverages deep residual features from the encoder while enabling precise spatial reconstruction through the decoder, making it well-suited for medical image segmentation.

3.1.6. Loss Function To address class imbalance and improve overlap quality in segmentation, a custom loss function was designed—Dice + Binary Cross Entropy (DiceBCELoss). Binary Cross Entropy (BCE) captures pixel-wise classification accuracy, while the Dice coefficient emphasizes accurate region overlap, which is particularly important for small anatomical structures. This combined loss enhances the model's capacity to correctly learn both the presence and shape of the lung regions.

3.1.7. Evaluation Metrics Model performance was quantitatively assessed using the following evaluation metrics:

- Accuracy
- Sensitivity (Recall)
- Specificity
- Dice Coefficient
- Jaccard Index (Intersection over Union, IoU)

These metrics collectively provide a detailed evaluation of segmentation quality, addressing both pixel-level correctness and region-level overlap.

3.1.8. Model Performance and Results The final model was selected based on the highest validation Dice coefficient, which occurred at Epoch 10. This checkpoint demonstrated strong generalization performance across multiple segmentation metrics.

These results indicate consistent performance across the training and validation datasets, with

Table 2 Performance Summary at Epoch 10

Metric	Training	Validation
Dice Coefficient	0.7727	**0.8521** (Best)
Jaccard Index (IoU)	0.6296	0.7423
Accuracy	0.8389	0.9058
Sensitivity (Recall)	0.9985	0.9985
Specificity	0.7786	0.8712
Combined Loss	-1.3446	-1.6483

particularly strong outcomes in sensitivity and segmentation overlap metrics, highlighting the model's robustness and generalization capability.

3.1.9. Observations and Insights The model demonstrated excellent foreground detection, achieving a sensitivity of 99.85%, which indicates highly reliable identification of lung regions with minimal false negatives. A high Dice coefficient (0.8521) and Jaccard Index (0.7423) reflect strong performance in terms of segmentation overlap and class balance, suggesting an effective balance between precision and recall. The validation accuracy of 90.58% highlights the model's strong generalization capability. Furthermore, the steady decrease in the combined loss (Dice + BCE) during training confirms stable convergence and effective learning behavior.

Conclusion Epoch 10 was selected as the final model due to its superior validation performance. It achieved high-quality segmentation with near-perfect sensitivity, balanced accuracy, and strong generalization. These results confirm the effectiveness of the proposed pipeline for downstream spatial or region-based tasks.

3.2. Pipeline B – Pathology Classification and Analysis

3.2.1. Preprocessing for Classification Before model development, the NIH ChestXray metadata was reviewed, and columns with predominantly missing values were removed to reduce noise and improve reliability. For image preprocessing, the same standardized steps from Pipeline A were applied to ensure consistency: grayscale conversion, resizing to 224×224 pixels, histogram equalization, and normalization to the [0, 1] range. This ensured clean, consistent inputs across both metadata and image modalities, supporting effective classification model development.

3.2.2. NIH ChestXray Labeling To generate spatially-aware supervision, the segmentation model from Pipeline A was applied to the entire NIH ChestXray dataset, producing lung region masks for each image. These binary masks highlighted pulmonary fields and served as intermediate annotations rather than direct inputs for classification. This approach enriched the dataset with pseudo-segmented regions, facilitating a weakly-supervised classification framework.

3.2.3. Postprocessing via PS-KDE To enhance the quality of the predicted masks and

reduce noise, a postprocessing step was introduced using **Probability Smoothing with Kernel Density Estimation (PS-KDE)**.

This method refines the raw segmentation masks by:

- Smoothing noisy pixel-level predictions
- Reinforcing consistent structural regions (e.g., lung contours)
- Eliminating spurious artifacts from the segmentation output
- Reducing image noise introduced by medical equipment or acquisition artifacts

The PS-KDE postprocessing was applied jointly on the original preprocessed images and their predicted lung masks, resulting in a new set of enhanced segmented images. These images maintain high anatomical fidelity while preserving essential pathological cues, and they serve as the final inputs for the classification model in **Pipeline B**.

This postprocessing step ensures that only high-confidence lung regions contribute to the learning process, improving model robustness, image quality, and interoperability.

3.2.4. Training Results and Model Performance The final model in **Pipeline A** was trained for up to 50 epochs, with early stopping triggered at epoch 15 due to stagnating validation performance. The best-performing checkpoint achieved the following metrics:

- **Training Loss:** 0.8479 **Validation Loss:** 2.1292
- **Training AUC:** 0.9024 **Validation AUC:** 0.7983
- **Training F1 Score:** 0.3056 **Validation F1 Score:** 0.2342
- **Training Accuracy:** 0.7693 **Validation Accuracy:** 0.7720
- **Training Exact Match Accuracy:** 0.0190 **Validation Exact Match Accuracy:** 0.0213

The model showed strong discriminative capability, as reflected in the high AUC scores on both training and validation sets. However, F1 scores and Exact Match Accuracy remained modest—primarily due to the challenges of multi-label classification and significant class imbalance.

Metric interpretation:

- **Accuracy** indicates the average proportion of correct labels per sample. It allows partial correctness (e.g., predicting 3 out of 5 labels yields 60% accuracy).
- **Exact Match Accuracy** is stricter: it requires an exact match between predicted and ground truth labels. Any deviation leads to a failed prediction, making this metric particularly difficult in the presence of label co-occurrence and imbalance.

Compared to earlier iterations that suffered from overfitting (high training scores but poor validation performance), this version showed improved generalization. Key improvements included:

- Targeted data augmentation strategies

- Use of an adaptive optimizer
- Class-frequency-based loss weighting to mitigate imbalance

Together, these enhancements yielded a more robust and stable model, establishing Pipeline A as a strong baseline and paving the way for the segmentation-aware classification in **Pipeline B**.

4. Discussion

PulmoTrack demonstrates strong performance across key tasks in lung imaging analysis using both CT and chest X-ray modalities. On the LUNA16 CT dataset, our unified 3D pipeline achieved competitive results in segmentation, detection, and malignancy classification of lung nodules. For chest X-ray analysis, the integration of PS-KDE preprocessing and a ResNet34-based U-Net led to high-precision segmentation of anatomical structures across JSRT, Montgomery, and NIH datasets. In addition, we successfully developed a multi-class classification model to detect multiple thoracic diseases from X-ray images, achieving strong performance aligned with state-of-the-art benchmarks. These results confirm the robustness of our approach. While each modality was handled through a distinct pipeline, both demonstrated generalization and diagnostic potential. Future integration of both modalities could further enhance clinical applicability.

5. Discussion

PulmoTrack demonstrates robust performance across key diagnostic tasks in lung imaging using both CT and chest X-ray modalities. On the LUNA16 dataset, our unified 3D pipeline achieved strong results in segmentation, detection, and malignancy classification of pulmonary nodules. For chest X-ray analysis, the integration of PS-KDE postprocessing and a ResNet34-based U-Net enabled high-precision segmentation of anatomical structures across JSRT, Montgomery, and NIH datasets. Furthermore, our multi-label classification model successfully detected a wide range of thoracic diseases from X-ray images, yielding performance aligned with or exceeding recent benchmarks.

These outcomes highlight the reliability and generalization capability of our dual-pipeline framework. While CT and X-ray data were processed through distinct architectures, both pipelines demonstrated clinical relevance and scalability. Notably, our approach achieved performance surpassing models reviewed in Gao et al. (3), especially on LUNA16.

Nonetheless, certain limitations remain—such as the challenge of segmenting very small nodules and the weak supervision inherent in some X-ray datasets. Future improvements may involve attention-based mechanisms, class-balancing strategies, and label refinement through semi-supervised learning. Additionally, integrating information

from both modalities through multimodal fusion could enhance diagnostic precision and adaptability to diverse clinical scenarios.

6. Conclusion

PulmoTrack introduces a dual-modality deep learning framework for comprehensive pulmonary analysis using CT and chest X-ray imaging. On the CT side, leveraging the LUNA16 dataset, our system performed accurate lung nodule segmentation, detection, and malignancy classification using 3D architectures. On the X-ray side, the combination of PS-KDE postprocessing and a ResNet34-based U-Net enabled high-fidelity anatomical segmentation, while a multi-label classification model effectively identified multiple thoracic pathologies.

Our segmentation models demonstrated strong generalization across multiple chest X-ray datasets (JSRT, Montgomery, NIH), often with minimal retraining. Importantly, the overall framework outperformed prior benchmarks in key evaluation tasks, particularly on the LUNA16 dataset, reinforcing its potential for AI-assisted pulmonary diagnosis.

In future work, we aim to explore multimodal fusion techniques to jointly exploit CT and X-ray data, while scaling evaluation to larger and more diverse patient cohorts. We also envision improvements in explainability and clinical integration to support real-world deployment in radiology workflows.

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